MILITARY SPECIFICATION

CARTRIDGES, 90MM HE-T, M71E1, AND 90MM SMOKE, WP-T, M313E1 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for two types of renovated cartridges designated as Cartridge, HE-T, M71E1 and Smoke, WP-T, M313E1 for use with 90MM Guns M36, M41 and M54.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-490 - Cleaning and Preparation of Ferrous and Zinc Coated Surfaces for Organic Protective Coatings.
MIL-C-20450 - Cases, Cartridge, Brass, Artillery, Resized.
MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.
MIL-T-11503 - Tracers, M5A2B1, M5A3B1, and T5E3, Loading, Assembling and Packing.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

DRAWINGS

ORDNANCE CORPS

XP-98203 - Cartridge, 90MM, HE-T, M71E1 and Cartridge, Smoke, WP-T, M313E1.

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7548306 - Box, Packing, AMM. for 90MM Gun AMM. in Fiber Container M53A3 Assembly, Details and Packing.
7548301 - AMM. Container, Fiber, M53A3.

PUBLICATIONS

ORDANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-6119339 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Cartridge.-The cartridge shall comply with all requirements specified on Drawing (dwg.) XP-98203 and with all requirements specified in applicable specifications.

3.3 Propellant weight.-The weight of the propellant shall not differ by more than plus or minus 0.25 ounce from that designated by the loading authorization for the particular lot of propellant used.

3.4 Cartridge case.-When the primer is removed from the case, the primer hole and counterbore along with mouth of case shall be gaged to assure compliance with the dwg. requirements.

3.4.1 Dents.-Dents in the case shall not exceed those permitted for resized cases in Specification MIL-C-20450.

3.5 Cartridge case crimp.-The cartridge case shall separate from the shell at a load between 5,000 and 11,000 pounds, when tested as specified in 4.3.1.

3.6 Tracers.-Records shall be furnished showing that the tracers used have been inspected and accepted in accordance with the criteria for M5A2B1 tracers of Specification MIL-T-11503 except as otherwise noted herein.

3.7 Proving ground test.-The cartridge shall comply with the following requirements, when tested as specified in 4.3.2.

3.7.1 Muzzle velocity.-The muzzle velocity at ambient temperature shall be not more than 2430 feet per second nor less than 2370 feet per second and the standard deviation shall not exceed 10.0 feet per second.

3.7.2 Tracer functioning.-The tracer shall ignite and there shall be visible evidence of the burning of the tracer composition.

3.7.3 Extraction.-The cartridge case shall extract freely from the gun after firing.

3.7.4 Safety.-There shall be no premature bursts or separation of metal parts in the gun bore or in flight before impact.

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3.8 Primers.-The primers shall have the required number of flash holes.

3.9 Workmanship.

3.9.1 General.-All parts shall be free from chips, dirt, grease, rust, burrs, and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. The cartridge or components shall not be damaged in any manner during the renovation or loading operation.

3.9.2 Painting and marking.-The painting and marking shall be restored by the loading and assembling contractor on any component furnished him when such painting or marking has been marred or obliterated.

3.9.3 Cartridge case.-The cartridge case shall be free of any roughness, scratches, rust or dents which will adversely affect their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring and testing equipment) and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 (except proving ground tests) and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

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4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agents authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied, and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records. (Specification MIL-G-2550 shall apply).

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4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product shall be called to the attention of the contractor immediately. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of cartridges of one designation produced by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting performance, safety or fit, as determined by the Government, shall not be cause for changing the lot number. Each lot shall contain:

a. Not more than one lot of previously acceptable complete rounds (prior to renovation).
b. Propellant from not more than one lot.
c. Loaded tracers from one manufacturer with one interfix lot number.
d. Loaded primers from one lot (applicable if primers are removed).
e. Cartridge cases from one manufacturer with one interfix lot number.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Case, prior to assembly of primer (see dwg. 7548413 covering a detail of dwg. XP-98203)

Categories	Defects	Method of inspection
Critical: None defined		
Major:	AQL 0.40 percent	
101.	Diameter of primer hole (see 3.4)	Gage
102.	Diameter of counterbore (see 3.4)	Gage
103.	Diameter of mouth (see 3.4)	Gage
104.	Depth of counterbore (see 3.4)	Gage

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Minor:	AQL 1.50 percent	
201.	Overall length, minimum (min.)	Gage
202.	Protective coating incomplete or damaged (see 3.9)	Visual
203.	Dents on case (see 3.4)	Visual
204.	Evidence of poor workmanship (see 3.9)	Visual

4.2.2.2 Primer, prior to assembly to cartridge case (see dwg. XP-98203)

Categories	Defects	Method of inspection
Critical:		
1.	Any flash hole missing	Visual

Major: None defined.

Minor: None defined.

4.2.2.3 Body, tracer (see dwg. BXP-96938 covering a detail of dwg. BXP-96937)

Categories	Defects	Method of inspection

Critical: None defined.

Major:	AQL 1.0 percent	
101.	Concentricity of thread with diameter of tracer cavity	Gage
102.	Pitch diameter of thread	Gage
103.	Major diameter of thread	Gage
104.	Diameter of tracer cavity, min.	Gage
105.	Diameter of counterbore, min.	Gage
106.	Depth of counterbore, min.	Gage
107.	Distance from closed end to end of thread, min.	Gage
108.	Distance from closed end to flange, maximum (max.)	Gage
109.	Distance from open end to shoulder	Gage
110.	Depth of tracer cavity, min.	Gage

Minor:	AQL 4.0 percent	
201.	Total length	Gage
202.	Outside diameter of closed end	Gage
203.	Outside diameter of cylindrical surface to rear of flange	Gage

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204.	Diameter of flange	Gage
205.	Distance from closed end to shoulder	Gage
206.	Distance from open end to flange	Gage
207.	Distance between wrench flats	Gage
208.	Profile of crimping lip improper	Gage
209.	Chamfer missing from mouth of tracer cavity	Visual
210.	Protective finish with bare spot	Visual
211.	Finish improper	Visual
212.	Evidence of poor workmanship (see 3.9)	Visual

4.2.2.4 Cartridge, prior to loading propellant (see dwg. XP-98203 and 3.7)

Categories	Defects	Method of inspection
Critical: None defined.		
Major: None defined.		
Minor:	AQL 1.5 percent	
201.	Case damaged (see 3.4 and 3.9)	Visual
202.	Foreign matter in interior of case	Visual
203.	Primer bent or otherwise damaged	Visual

4.2.2.5 Shell loading assembly, prior to assembly of tracer (see dwg. XP-96928 or dwg. XP-96878 covering a detail of dwg. XP-98203)

Categories	Defects	Method of inspection
Critical:		
1.	Depth of tracer cavity, maximum (max.)	Gage
Major: None defined.		
Minor:	AQL 2.50 percent	
201.	Pitch diameter of tracer thread, max.	Gage
202.	Minor diameter of tracer thread, max.	Gage
203.	Length of perfect tracer thread, max.	Gage
204.	Depth of tracer cavity, min.	Gage
205.	Evidence of poor workmanship (see 3.9)	Visual

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4.2.2.6 Tracer assembly, prior to assembly to shell (see dwg. XP-96937 covering a detail of dwg. XP-98203)

Categories	Defects	Method of inspection
Critical: None defined.		
Major: None defined.		
Minor:	AQL 0.40 percent	
201.	Foreign matter (see 3.9)	Visual

4.2.2.7 Shell loading assembly (see dwg. XP-96785 or dwg. XP-98444 covering a detail of dwg. XP-98203)

Categories	Defects	Method of inspection
Critical: None defined.		
Major:	AQL 0.25 percent	
101.	Gasket missing	Visual
Minor:	AQL 1.00 percent	
201.	Tracer turns when minimum torque is applied	Test
202.	Evidence of poor workmanship (see 3.9)	Visual

4.2.2.8 Cartridge (see dwg. XP-98203)

Categories	Defects	Method of inspection
Critical:		
1.	Primer above flush	Gage
Major:	AQL 0.25 percent	
101.	Assembly damaged	Visual
Minor:	AQL 1.50 percent	
201.	Fuze turns when minimum torque is applied	Test
202.	Exterior coating damaged	Visual
203.	Marking misleading or unidentifiable	Visual
204.	Evidence of poor workmanship (see 3.9)	Visual

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4.2.2.9 Unsealed fiber container (see dwg. 7548301)

Categories	Defects	Method of inspection
Critical: None defined.		
Major:	AQL 0.25 percent	
101.	Packing component missing	Visual
Minor:	AQL 1.50 percent	
201.	Container cap not removable by hand	Manual
202.	Cartridge not removable by hand	Manual
203.	Cartridge incorrectly assembled in container	Visual
204.	Evidence of poor workmanship (see 3.9)	Visual

4.2.2.10 Sealed fiber container (see dwg. 7548301)

Categories	Defects	Method of inspection
Critical: None defined.		
Major:	AQL 0.40 percent	
101.	Tape incomplete or badly wrinkled	Visual
102.	Container cut or damaged through all impregnated layers	Visual
Minor:	AQL 1.00 percent	
201.	Contents loose	Manual
202.	Marking misleading or unidentifiable	Visual

4.2.2.11 Sealed wood packing box (see dwg. 7548306)

Categories	Defects	Method of inspection
Critical: None defined.		
Major:	AQL 0.40 percent	
101.	Box damaged	Visual
102.	Strapping missing, broken or loose	Visual
103.	Boards loose or nails protruding	Visual
Minor:	AQL 0.40 percent	
201.	Marking misleading or unidentifiable	Visual

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4.2.3 Testing

4.2.3.1 Check test for deterioration of primers - Major defect.-If the total time between original acceptance of any primer lot and the assembly of that lot into complete round cartridges extends two years, or if the primers have been subject to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to, and must satisfactorily pass the check test, specified in the applicable prin. " loading specification immediately before the primer lot is assembled into the cartridges. The test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into the cartridges.

4.2.3.2 Freedom from alkalies and strong acids (see dwg. XP-98203 and Specification MIL-C-490) Minor defect.-When the protective coating is removed from the case or shell assembly by a chemical process, three samples shall be selected from each eight hours of production for each test. If any case or shell assembly fails to meet the applicable requirement, it shall be rejected and action shall be taken by the contractor to correct the cause of failure. The tests shall be performed as specified in Specification MIL-C-490.

4.2.3.3 Propellant weight (see 3.3) Critical defect.-The propellant shall be weighed and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing, using a different balance from that used to conduct the original weighing and, if performed manually, shall be performed by another operator. Any charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accorda with 4.2.4.

4.2.3.4 Cartridge case crimp (see 3.5) Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-414, Section B, Table B-3, Code letter F and AQL 1.5 percent. The test shall be performed as specified in 4.3.1, with equipment in accordance with 4.2.4 (see 6.4).

4.2.3.5 Proving ground.-The proving ground test shall be performed as specified in 4.3.2 (see 6.4).

4.2.3.5.1 Muzzle velocity (see 3.7.1) Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-414, Section B, Table B-3, Code letter D and AQL 1.0 percent. In addition, if the standard deviation of the velocity of the sample multiplied by the factor 0.65 exceeds the applicable requirement for standard deviation, the lot shall be rejected.

4.2.3.5.2 Tracer functioning (see 3.7.2) Major defect.-The sample . rtridges fired in the test for muzzle velocity shall be observed for tracer functioning. If two or more tracers fail to comply with the applicable requirement, the lot shall be rejected.

4.2.3.5.3 Safety (see 3.7.4) Critical defect.-A sample of 5 cartridges shall be selected for this test. If one or more rounds fail to comply with the applicable requirements, the lot shall be rejected.

4.2.3.6 Profile and alignment (see dwg. XP-98203) Major defect.-The cartridges shall be gaged 100 percent for profile and alignment. Any cartridge which fails to meet the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

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4.2.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.2.2 and 4.2.3, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring and testing equipment identified by Index of Inspection Equipment No. 6119339, or approved equivalents thereto. In inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.

4.3.1 Cartridge case crimp.-The security of the cartridge case crimp to the shell body shall be determined by applying a load in the axial direction at the rate of 0.125 inch per minute using test equipment in accordance with 4.2.4. Observation shall be made for compliance with the applicable requirement.

4.3.2 Proving ground test.

4.3.2.1 Muzzle velocity and tracer functioning.-This test shall be conducted at a Government proving ground in accordance with the latest Ordnance proof manual for the round undergoing test. Observation shall be made for compliance with the applicable requirements.

4.3.2.2 Safety.-Assemble the rounds with a charge adjusted to give pressures 112 percent of the average pressure for the particular lot of propellant. The test shall be conducted as specified in 4.3.2.1 and observation shall be made for compliance with the applicable requirement.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-The cartridge shall be preserved and packaged in accordance with dwg. 7548301.

5.2 Packing.

5.2.1 Level A.-The container shall be packed in accordance with dwg. 7548306.

5.3 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

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6.2 Check test for deterioration of primers.-The contractor shall be reimbursed for the expenses incurred in the performance of check tests, upon approval from the procuring activity. The test shall be conducted at Government expense, without cost to the contractor assembling primers into the cartridges. This test shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.3 Intended use.-The cartridges procured under this specification are intended to be produced from previously acceptable rounds, prior to modification.

6.4 Submission of test data.-In addition to the normal distribution or records, when the cartridge is procured by the Department of the Army, one copy should be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: ORDBB-NR6

6.5 Drilling of tracer cavity.-It is recommended that the drilling of the tracer cavity for the tracer body in the shell loading assembly (Comp B loaded) be accomplished without exceeding a metal temperature (at the inside surface of shell base) of 75°Centigrade.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army-Ordnance Corps

Preparing Activity:
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